

Lab 1 - RedFlag Descriptive Paper

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1 Introduction

The rise of digital communication and social media platforms has significantly changed how people consume information. News, opinions, and current world events can now be shared instantly across platforms such as Instagram, X (formerly Twitter), and online news websites. While this accessibility has increased the speed of information distribution, it has also created an environment where misinformation can spread rapidly and influence public understanding. Misinformation refers to false or misleading information that may be unintentionally or intentionally shared, often without verification. This issue has become a growing societal problem because users are constantly exposed to large volumes of content and may not have the time or resources to verify every claim they encounter.

The spread of misinformation has been studied extensively in recent years, particularly on social media platforms. Research conducted by Massachusetts Institute of Technology found that false news spreads significantly faster and farther than truthful news online (Vosoughi et al., 2018). The study analyzed over 126,000 stories shared by approximately 3 million users and concluded that false information was 70 percent more likely to be reshared than truthful information (Vosoughi et al., 2018). Truthful stories took approximately six times longer to reach the same audience size compared to false stories (Vosoughi et al., 2018). These findings demonstrate how misinformation can gain momentum far more quickly than corrections can follow, making it difficult for users to identify credible information before it spreads widely.

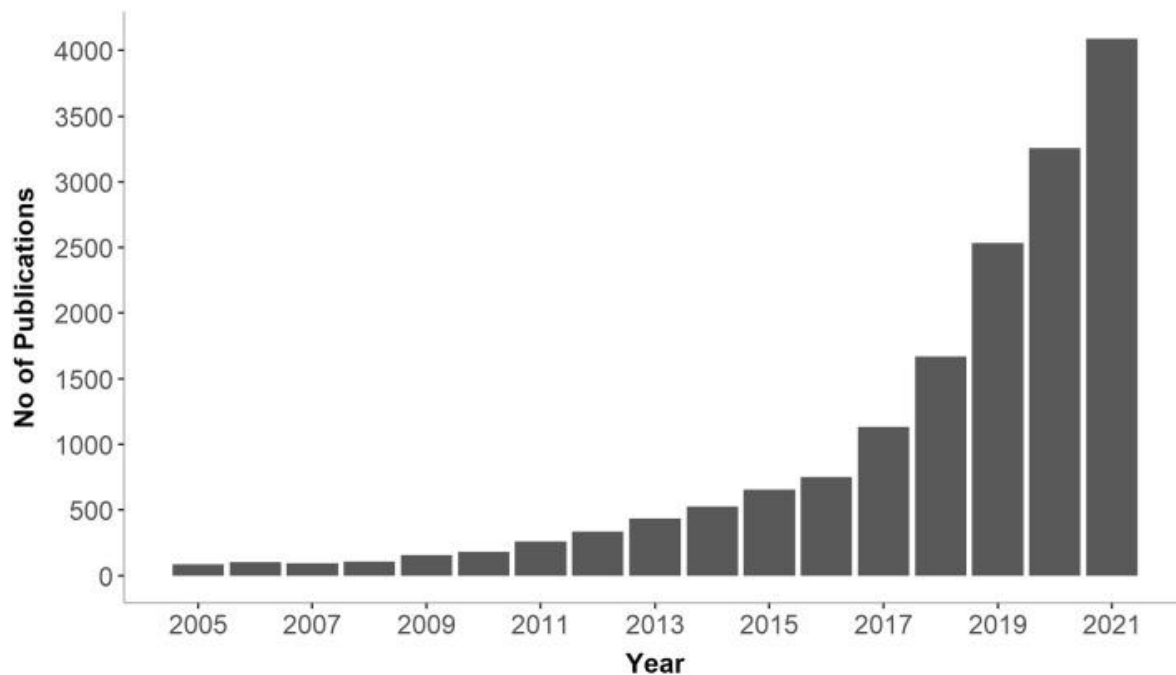
Current solutions for combating misinformation are often limited in accessibility and efficiency. Most fact-checking systems require users to manually search for information using external websites such as Snopes or PolitiFact. While these resources are effective, they interrupt

the user's browsing experience and create friction that discourages consistent use. In many cases, users must leave the content they were viewing, search for verification, compare sources, and then interpret findings independently.

Another limitation in current misinformation solutions is the lack of contextual bias awareness. Political and social issues are often presented through biased viewpoints, reinforcing echo chambers where users are exposed to viewpoints that are similar to their own. This ends up creating filter bubbles, which limit critical analysis and balanced understanding of important issues (Pariser, 2011). Existing systems generally focus on fact verification but do not provide users with alternative perspectives that may improve informed decision making.

Figure 1

Graph showing misinformation growth over time



To address these gaps, a solution must be passive and immediate, integrated directly into the user's browsing experience. RedFlag is designed as a real-time misinformation detection and bias awareness tool that addresses these needs. It operates as a background browser extension that scans content as users browse the internet, identifies potentially misleading information, and flags it on the screen. It also introduces BiasStar, a balanced perspectives engine that identifies politically divisive content and presents sources from both sides of the political spectrum. By combining real-time fact checking and bias awareness, RedFlag aims to improve and support more informed decision-making in online environments.

2 Product Description

RedFlag is a fact-checking and bias indicator designed to operate across multiple platforms. It runs in the background, analyzing text to find any potentially inaccurate information and provide relevant sources for the user to consult. By providing sources for the user, RedFlag allows the user a way to look further into the misinformation. RedFlag's BiasStar feature adds a star to politically divisive content and offers further viewpoints from sources across the political spectrum.

2.1 Key Product Features and Capabilities

RedFlag comes with several features and capabilities to set it apart from other fact-checking tools. It comes with real-time content detection: it scans the user's screen to find any misinformation, working automatically in the background without requiring any user input. It addresses the inconvenience of the user having to leave their current browsing experience by bringing sources directly to them.

RedFlag can highlight any questionable information on the user's browsing experience with no input, not intruding on the user's experience and presents the highlight subtly. It intends to intervene before the user potentially shares misinformation, while not entirely interrupting the experience.

RedFlag also comes with a balanced perspective engine, called BiasStar, that identifies political content and provides multiple viewpoint sources across the political spectrum. It does not label ideologies explicitly but does offer an evaluation from every perspective. It addresses how social media may reinforce bias and echo chambers.

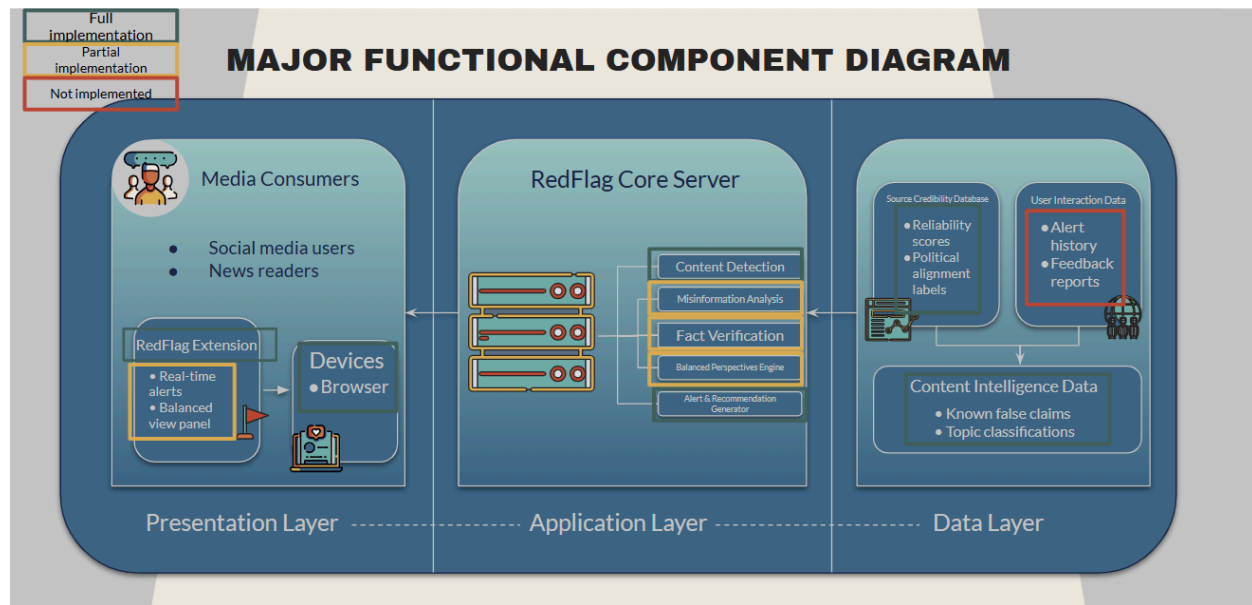
RedFlag users can also get on-demand explanations on why the information they were seeing was flagged. This feature allows the user to make a more informed decision without forcing how the user should think about the topic, providing better credibility towards content and allowing the user to think further on the information provided.

2.2 Major Components (Hardware/Software)

RedFlag mainly relies on the user's software, only requiring a system capable of running a modern web browser. The system is made to run on common browsers, such as Google Chrome and Microsoft Edge, and operating systems such as Windows and macOS. RedFlag is made to be accessible to any user.

Figure 2

Major Functional Component Diagram



RedFlag's presentation layer consists of the RedFlag browser extension and is both the front-end interface, and is responsible for displaying the real-time alerts and a view panel to users. This layer provides the primary interface through which users interact with RedFlag while browsing. It is built with JavaScript, allowing it to run on browsers with ease.

RedFlag's application layer contains the primary processing components of RedFlag, including content detection, misinformation analysis, fact verification, the Alert and Recommendation generator and the Balanced Perspectives Engine. These components all work together to analyze content and determine what information should be presented to the user. The backend uses Python with Flask/Django frameworks to process information and interact with the project's information datasets.

The data layer consists of a PostgreSQL database which is used to store the information required by RedFlag's various systems. It includes the Source Credibility Database, which is responsible for containing the reliability scores and political alignments, and User Interaction data and Content Intelligence data. The Content Intelligence data keeps the known false claims

and topic classifications, allowing the application layer to compare newly detected information against previous information data.

RedFlag uses various External Interfaces and APIs to supplement its various systems. The fact-checking API compares questionable information against any available databases, while the News API takes publicly available news as input then categorizes it by topic. A large language model (LLM) API is used to parse database information and summarize its findings, which in turn helps increase the speed at which RedFlag can perform fact-checking and present information to the user.

3 Identification of Case Study

Daiskue is a 16-year-old teenager who frequently uses Instagram during his free time. Although he recognizes that some of the information appearing on his Instagram feed may be false, he does not want to spend the time manually researching every post he encounters. RedFlag addresses this problem by providing misinformation indicators that allow users to identify potentially questionable information without having to independently research every post. When Daiskue enables RedFlag through his browser, the system monitors the content he encounters and displays indicators wherever questionable text is detected. These indicators provide Daiskue with multiple sources, allowing for further investigation without having to navigate away. RedFlag also provides additional capabilities through features such as highlighting warnings, real-time integrated detection, on-demand explanations, and BiasStar. This case study focuses on users like Daiskue who regularly encounter large amounts of online information and may benefit from assistance evaluating its credibility. As RedFlag becomes more widely known, Daiskue could introduce RedFlag to his friends, other students, and general

social media users who encounter potentially misleading information and want additional context while browsing.

4 Glossary

BiasStar: RedFlag's internal system for identifying politically divisive content

Credibility Score: A score given to a claim that is found online rating the credibility of the claim, or the likelihood that the claim is correct.

Disinformation: False or inaccurate information that is shared with the intention to mislead others

End-user: The person who uses or is intended to use a computer system or other software application

Misinformation: False or misleading information shared regardless of intent

5 References

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